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Effect of microaeration combined with ferric chloride pretreatment on anaerobic digestion of corn stalk

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ABSTRACT

In this study, microaeration combined with ferric chloride pretreatment was used to improve the methanogenic performance for thermophilic dry anaerobic digestion. Parameter optimization and mechanism analysis were explored for combined pretreatment in anaerobic digestion by response surface methodology and microbial diversity and community. The results showed that both microaeration and ferric chloride had significant effects on the cumulative methane production, and the effect of microaeration was greater than that of ferric chloride. The optimal pretreatment parameters were predicted at oxygen loading rate of 0.76 mL/g VS and ferric chloride addition rate of 4.04 mg/g VS. The methane production of anaerobic digestion was 114.05 mL/g VS with the optimal pretreatment parameters, which was 17.78% higher than that without pretreatment. In addition, microbial community analysis showed that combined pretreatment enhanced methanogenesis by enhancing the hydrogenotrophic pathway.

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Anaerobic digestion; pretreatment; methane production; microbial community; corn stalk

Introduction

Anaerobic digestion is a green biotechnology that converts organic matter from waste into methane (Wang, Li, and Yu et al. 2022). At present, there are mainly three categories of wet, semi-dry, and dry anaerobic digestion, whose TS is ≤10%, 10–15%, and ≥15%, respectively (Zhen, Luo, and Xing et al. 2021). Among them, semi-dry or dry anaerobic digestion has not been widely promoted due to the lack of adequate technology to keep the digestive system running steadily compared to wet anaerobic digestion (Deepanraj, Senthilkumar, and Ranjitha 2021). Some problems such as poor fluidity and mass transfer of feedstock and low stability of the digestive system remain to be solved in dry anaerobic digestion, which leads to excessive accumulation of volatile fatty acids (VFAs) or ammonia nitrogen (Esen and Yuksel 2013; Rocamora, Wagland, and Villa et al. 2020). However, greater TS, less digestate generation, smaller digesters, and higher biogas production make semi-dry and dry anaerobic digestion more attractive.

In view of the existing problems of dry anaerobic digestion, several methods, such as temperature adjustment, pretreatment, and additives, have been tried from the two main processes of anaerobic digestion, namely, hydrolytic acidification and methane production (Kainthola, Kalamdhad, and Goud 2019b; Sun, Wang, and Liu et al. 2020a). In general, microbial activity is stimulated through ambient temperature, thereby shortening the digestion cycle and increasing the organic load rate (Nie, He, and Zhang et al. 2021). Some facultative bacteria that release proteolytic and cellulolytic enzymes are propagated by microaeration (MA) pretreatment, which can improve the hydrolytic acidification process (Amin, Khalid, and Li et al. 2021). The addition of trace elements is also a convenient way to adjust the stability of dry anaerobic digestion, which can accelerate the growth of methanogens (Yu,

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Zhang, and Dai et al. 2016). Some trace elements (such as Fe, Co, and Ni) have been shown to be essential nutrients needed for the growth of methanogens (e.g. methyl-coenzyme M and coenzyme M methyl-transferase) (Zhang, Wu, and Guo et al. 2015). Moreover, Fe is one of the most needed metal elements for microorganisms, which can accept/donate electrons in the redox reaction, so as to improve microbial activity and accelerate substrate hydrolysis (Ugwu and Enweremadu 2020). However, a single potential improvement method often has bottlenecks and limitations that are difficult to break. It seems feasible to integrate multiple processes to optimize the performance of dry anaerobic digestion.

Conventionally, "one at a time" variation of variables is applied to the optimization of many industrial processes, in which the response variable is only a function of a single variable parameter, and the processing parameters are assumed to be non-interacting (Ahmadi, Vahabzadeh, and Bonakdarpour et al. 2005). When multiple independent variables interact together to affect the desired response, the response surface methodology (RSM) can effectively optimize and evaluate the process (Kainthola, Kalamdhad, and Goud 2019a). Therefore, RSM provides a set of analysis and solution programs for many industrial process problems, such as optimal product design, process improvement, and system optimization (Gunes, Stokes, and Davis et al. 2021).

Based on the above analysis, the effects of microaeration combined with ferric chloride (MAFC) pretreatment on thermophilic anaerobic digestion of corn stalk (CS) were investigated. The specific purposes of this study were: (1) To optimize the optimal conditions of MAFC pretreatment and explore the effect of pretreatment conditions on cumulative methane production (CMP) by central composite design (CCD); (2) To reveal the enhancement mechanism of MAFC pretreatment on anaerobic digestion performance through microbial diversity and community.

Materials and Methods

Substrate and inoculum

The characteristics of CS and inoculum are listed in Table 1. CS was collected from a farm in Heilongjiang, China, which was air-dried and crushed to nearly 1 cm by a pulverizer, and then stored at room temperature. The inoculum was derived from a 10 L mesophilic anaerobic digester.

Experimental design

Central composite design

The CCD is used to convert two independent variables into dimensionless variables (x_1, x_2), whose coded values are of five levels: −a, −1, 0, +1, +a. Based on the results of the pre-experiment, the actual representation and magnitude order for the coded value were assigned and listed in Table 2. According to the CCD arrangement in Table 3, the experiment required 13 runs, and all experimental and predicted values are listed. Then, the evaluation of lack of fitness and experimental error was carried out through the central operation.

Table 1. Characteristics of the CS and inoculum.

Parameters	Corn Stalk	Inoculum
TS (%)	94.12 ± 0.94	1.76 ± 0.07
VS (% of TS)	87.31 ± 1.35	0.85 ± 0.03
pH	/	7.29 ± 0.01
C (% of TS)	41.14 ± 0.17	/
N (% of TS)	0.61 ± 0.02	/
Ash (% of TS)	6.81 ± 0.08	/
Hemicellulose	27.28 ± 3.12	/
Cellulose	36.42 ± 4.31	/
Lignin	9.45 ± 1.52	/

Notes: /, not determined.

Table 2. Experimental factors and levels for the central composite design.

Independent Variable		Units	Real values of coded levels				
			−1.414	−1	0	+1	+1.414
x_1	Oxygen load	mL/g VS	0	0.44	1.5	2.56	3
x_2	FC ratio	mg/g VS	0	0.88	3	5.12	6

Table 3. Arrangement of the CCD and experimental/predicted values for CMP.

Run	Code values		Real values		Experimental CMP	Predicted CMP
	x_1	x_2	Oxygen load	FC ratio		
1	0	1.414	1.5	6	106.54	109.28
2	−1	−1	0.44	0.88	106.43	106.41
3	−1	1	0.44	5.12	115.92	113.20
4	0	0	1.5	3	113.34	112.08
5	0	0	1.5	3	113.2	112.08
6	1.414	0	3	3	99.09	99.16
7	1	−1	2.56	0.88	98.45	99.51
8	0	−1.414	1.5	0	103.21	102.13
9	1	1	2.56	5.12	104.47	102.83
10	0	0	1.5	3	112.04	112.08
11	0	0	1.5	3	110.47	112.08
12	0	0	1.5	3	111.34	112.08
13	−1.414	0	0	3	109.78	111.37

In order to confirm the rationality of the hypothesis, independent variables and their interactions were evaluated with an estimate of 0.05 confidence after completion of the experiment. The normality of the data points was also proved by the normal plot of residual. Response (CMP) was correlated to actual value (oxygen load and ferric chloride ratio) using the appropriate Eq. (1):

$$y = \alpha_0 + \alpha_1 x_1 + \alpha_2 x_2 + \alpha_{12} x_1 x_2 + \alpha_{11} x_1^2 + \alpha_{22} x_2^2 \quad (1)$$

where y is the dependent variable (CMP, mL/g VS); α_0, α_1, and α_2 are coefficients of linear expressions; α_{11} and α_{22} are the quadratic coefficients; α_{12} is the interaction coefficients; x_1 and x_2 are the coded experiment factors, namely, oxygen load (mL/g VS) and ferric chloride ratio (mg/g VS).

MAFC pretreatment

According to the CCD arrangement, different conditions of MAFC pretreatment were satisfied. Firstly, 50 g CS, 75 mL deionized water, and 200 mL inocula were mixed in each 500 mL serum vial. Then, different proportions of ferric chloride (FC) reagents were added to the serum vial. FC reagents of 0, 0.88, 3, 5.12, and 6 mg/g VS were added to different serum vials to form different mass ratios of FC and CS. A certain proportion of oxygen (0, 0.44, 1.5, 2.56, and 3 mL/g VS) was slowly injected into the bottom of different serum vials through a syringe during the anaerobic digestion process each day to form a microoxygen environment with different loads.

Anaerobic digestion experiments

The anaerobic digestion experiment was divided into two parts. The first part was used to evaluate the CMP of different MAFC pretreatments in an automated biomethane potential test system with 15 channel reactors (AMPTSII, Bioprocess Control, Lund, Sweden). The system consists of a digestion unit, a CO_2 removal unit, and a gas flow measurement unit, which enables on-line monitoring of biomethane. All serum vials were digested at 36°C for 25 days with a blender at 50 rpm.

The second part is to explore the potential mechanism of MAFC pretreatment to improve the anaerobic digestion performance. Firstly, the optimal pretreatment conditions should be determined based on the CMP results of anaerobic digestion and the optimization of RSM. Then, the two serum

vials were retested for anaerobic digestion to compare and analyze performance differences. One was pretreated with optimum MAFC conditions and the other was not pretreated (CK). In addition, daily methane production was monitored to assess the biomethane performance of anaerobic digestion. Digested substrates were collected at the beginning, middle, and end of anaerobic digestion for the impact of MAFC on the microbial community by 16S rRNA sequencing. Three parallel replicates were performed to ensure repeatability of the experiment.

Analytical Methods

The APHA standard method was used for TS and VS in the samples. The lignocellulose component contents (hemicellulose, cellulose, and lignin) were measured according to the Van Soest method. The C and N contents of the samples were analyzed by the elemental analyzer (EA 3000, LEEMAN Technologies Co., Ltd., Beijing, China). The experimental data was analyzed by Origin 2017 (Origin Lab Corporation, Northampton, MA, USA), and the RSM was done on Design Expert 10.0 (Stat-Ease Inc., US).

Microbial diversity and community

The microbial community was analyzed using 16S rRNA sequencing technology. To analyze the system state of anaerobic digestion, digestive samples were collected for DNA extraction on day 5, 15 and 25, respectively. The universal primers were used to amplify the bacterial 16S rRNA sequence fragment. Sequencing was analyzed on the Illumina HiSeq 2000 platform (Illumina, San Diego, USA). Operational taxonomic unit (OUT) clustering was analyzed by Ribosomal Database Project (RDP) database (http://rdp.cme.msu.edu/).

Calculation Methods

A first-order kinetic model was applied to fit the hydrolytic rate k (day^{-1}) of the specific methane production, as shown in Eq. (2):

$$B(t) = B_m \times \left(1 - e^{-kt}\right) \tag{2}$$

where B(t) is the specific methane production (mL/g VS), B_m represents the simulative specific methane production (mL /g VS), and k (day^{-1}) represents the hydrolytic rate.

A modified Gompertz model was applied to determine the simulative CMP, maximum methane production potential, and lag phase for anaerobic digestion, and is shown as following:

$$G(t) = G_m \times exp\left\{-exp\left[\frac{r_m \times e}{G_m}(\lambda - t) + 1\right]\right\} \tag{3}$$

where $G(t)$ is the CMP (mL/g VS), G_m represents the simulative specific methane production (mL/g VS), r_m is the maximum rate of methane production (mL/g VS/day), λ represents the lag phase (day), while t is the digestive time.

Results and discussion

Diagnostic checking and statistical analysis of the model

In this study, the effect of MAFC pretreatment parameters (oxygen load and FC ratio) on CMP was investigated by CCD. After applying several regression investigations, a second-order polynomial equation representing the correlation between the two pretreatment parameters and CMP in was simulated by multiple regression analysis of the experimental data. Therefore, the regression model in response to CMP based on coded and uncoded factors was as follows

Table 4. Analysis of variance for the response variables.

Source	Sum of Squares	df	Mean Square	*F* Value	*p*-Value Prob > *F*	
Model	337.20	5	67.44	16.61	0.0009	Significant
x_1-Oxygen load	149.20	1	149.20	36.75	0.0005	
x_2-FC ratio	51.10	1	51.10	12.59	0.0094	
$x_1 x_2$	3.01	1	3.01	0.74	0.4177	
x_1^2	80.70	1	80.70	19.88	0.0029	
x_2^2	70.61	1	70.61	17.39	0.0042	
Residual	28.42	7	4.06			
Lack of Fit	22.43	3	7.48	5.00	0.0770	Not significant
Pure Error	5.98	4	1.50			
Cor Total	365.61	12				

$$y = 112.08 - 4.32x_1 + 2.53x_2 - 0.87x_1x_2 - 3.41x_1^2 - 3.19x_2^2 \quad (4)$$

Or

$$y = 99.69 + 6.18x_1 + 6.02x_2 - 0.38x_1x_2 - 3.03x_1^2 - 0.71x_2^2 \quad (5)$$

A good statistical model should have an R^2 value in the range of 0.75–1, which will guide the best fit of the model (Kainthola, Kalamdhad, and Goud 2019a). The R^2 of this model was 0.9223, indicating that 92.23% variability in the CMP response can be explained. Statistical parameters obtained from the analysis of variance (ANOVA) for the adequacy and significance of the regression model are given in Table 4. The *F* value and *p*-value of the quadratic model were 16.61 and 0.0009 (<0.01), respectively, which meant the quadratic model was significant. The lack of fit value was 22.43 (p = 0.0770, not significant), implying that it was significantly relative to the pure error. Additionally, the effect of MA on CMP was greater than that of FC due to the larger F value (Wang, Cheng, and Liu et al. 2019a), and they were significant for CMP ($p < .05$).

To check the adequacy of the model, diagnostic model items were calculated, and standardized residuals showed slight differences between the hypothetical model items and the response model items. In Figures S1, normal probability plots of standardized residuals and actual and predicted CMP curves were shown. The more balls are distributed on the slash, the better the confidence is shown for the model. The results showed that there was no anomaly in the experiment, and the expectation of CMP enhancement was obtained through the quadratic model.

Effect of MAFC pretreatment on CMP

As shown in Figure 1(a), when the addition ratio of FC was 3 mg/g VS (central level), CMP gradually increased with microaeration pretreatment. However, a downward trend was shown in the CMP at oxygen loading rates in excess of 0.43–0.86 mL/g VS. This result was consistent with studies by Fu, Wang, and Yuan et al. (2015) who showed that moderate aeration could improve CMP. They confirmed that the CMP for anaerobic digestion of CS was increased by 16.2% at oxygen load of 5 mL/g VS. The microoxygen environment is conducive to the metabolism of aerobic and facultative anaerobic microorganisms, which increases the content of cellulose, protein, and other hydrolytic enzymes to promote the hydrolysis of complex organic compounds (Brémond, de Buyer, and Steyer et al. 2018). However, the activity of anaerobic methanogens is inhibited by toxicity at higher oxygen levels, which will compete with facultative microorganisms on the substrates (Xu, Selvam, and Wong 2014). Thus, excessive aeration has a negative effect on CMP during anaerobic digestion (Lim and Wang 2013). In addition, rapid hydrolysis of substrates may lead to excessive accumulation of VFA, resulting in inhibition of anaerobic digestion (Sheets, Ge, and Li 2015). Zhen, Luo, and Xing et al. (2021) found that the increased butyric acid content exceeded 8 g/L with the excess oxygen load, which led to the inhibition of dry anaerobic digestion.

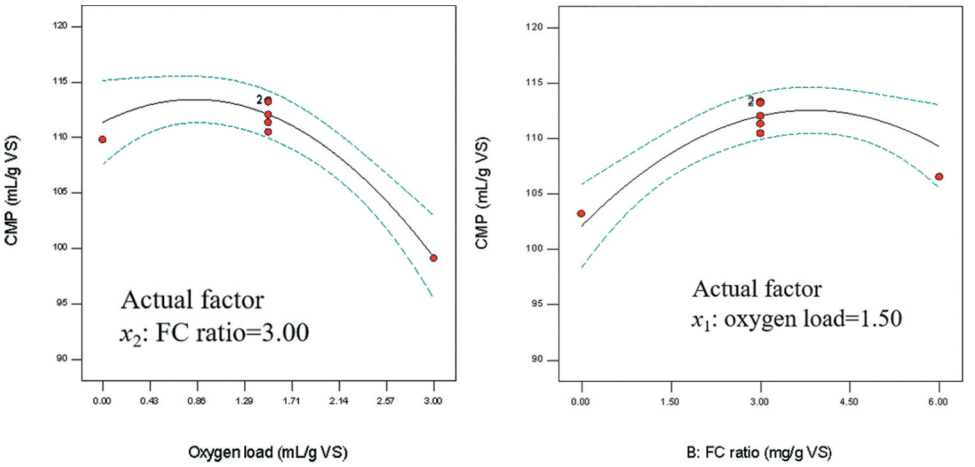


Figure 1. Single factor impact analysis. Effect of (a) oxygen load and (b) FC ratio on CMP.

Figure 1(b) shows that a higher FC addition rate resulted in a higher CMP (Oxygen load was retained at its central level). This is due to the fact that FC can be used as an electron acceptor to accelerate complex organic hydrolysis and promote the bioconversion of reducing sugars and amino acids to acetic acid by the acquisition of intermediate electrons (Zhan, Tian, and Zhang et al. 2021). Zhao, Li, and Zhao et al. (2020) also reported that dissimilated iron reduction was the primary route to improve the methanogenic performance of FC pretreatment. The Fe(III)-reducing microorganisms could be enriched by the addition of ferric chloride and the use of Fe(III) as an electron acceptor to promote organic decomposition via the DIR process (Baek, Kim, and Lee 2014). Another explanation is that the supplementation of ferric chloride can enrich *Coprothermobacter* for proteins fermentation and *Methanosarcina* for methanogenesis (Yu, Lou, and Zhang et al. 2015a). However, the limited increase in the CMP even began to be inhibited as the FC reached 3–4.5 mg/g VS. This result was similar to the study of Zhan, Tian, and Zhang et al. (2021), and the FC dosage below 390 mg Fe(III)/L would not bring a huge shock to the anaerobic digestive. Due to excessive FC supplementation, pH decreased sharply along with the rapid hydrolysis of FC (Eq. (6)), which may impair methanogen activity and thus give rise to a poor CMP (Qin, Chen, and Wang et al. 2019).

$$Fe^{3+} + 3H_2O \rightarrow Fe(OH)_3 \downarrow + 3H^+ \tag{6}$$

In summary, restricted oxygen increased the hydrolysis rate of the substrate, but it also faced the potential problem of excessive accumulation caused by rapid VFA generation. On the other hand, although high Fe concentrations caused digestive system collapse, appropriate FC supplementation accelerated VFA conversion and promoted methanogen activity, which prevented excessive accumulation of VFA formed by rapid substrate hydrolysis and improved methane production efficiency. Therefore, a high CMP could be obtained by MAFC pretreatment. From Figure 2, the 3D model showed that the maximum predicted CMP was obtained at oxygen loading rate of 0.76 mL/g VS and ferric chloride addition rate of 4.04 mg/g VS, which can be used as optimized pretreatment conditions for subsequent studies.

Methane production performance during dry anaerobic digestion

Figure 3(a) shows the daily methane production of anaerobic digestion for raw and MAFC pretreated CS. CK obtained methane production of 2.64 mL/g VS, which was 105.47% higher than that of the MAFC pretreatment on the first day. As the anaerobic environment was destroyed by MAFC pretreatment, the activity of methanogens was inhibited to a certain extent. Afterward, daily methane

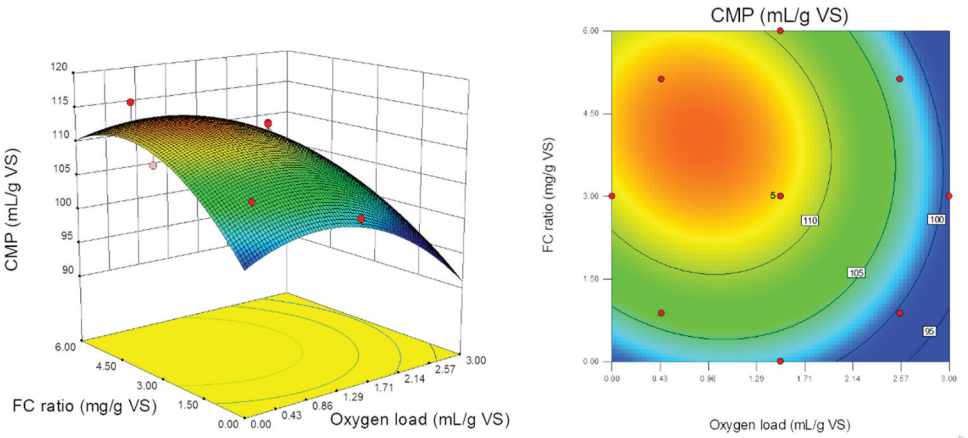


Figure 2. (a) Response surface and (b) contour plots of different parameters on CMP.

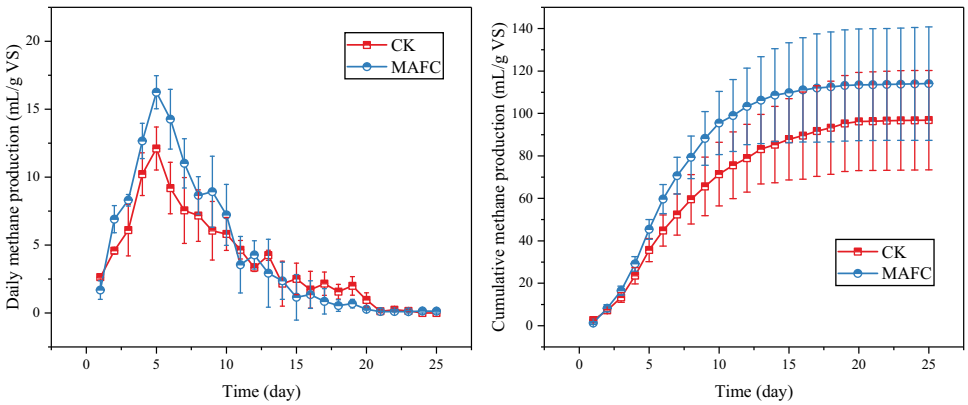


Figure 3. Methane production properties in anaerobic digestion. (a) Daily methane production and (b) cumulative methane production.

production for MAFC pretreatment gradually increased and exceeded the daily methane production for CK by day 3, which can be attributed to the fast hydrolysis, acidogenesis, and methanogenesis (Chen, Yang, and Zhao et al. 2019a). In the following days, daily methane production remained at a high level in both groups and peaked on the fifth day. A maximum daily methane production of 16.25 mL/g VS was obtained for MAFC, which was 34.19% higher than that for CK (12.11 mL/g VS). In addition, after 13 days of anaerobic digestion, the daily methane production of CK was again higher than that of MAFC, due to the accelerated hydrolysis rate and rapid substrate depletion caused by MAFC pretreatment. Eventually, the CMP detected was 114.05 mL/g VS for MAFC, exceeding 17.78% of CMP for CK (Figure 3(b)).

In order to compare the hydrolytic and methanogenic performance of anaerobic digestion before and after MAFC, first-order kinetic model and modified Gompertz model were applied to fit the profiles of CMP (Xu, Awasthi, and Li et al. 2020). The high coefficient ($R^2 > 0.945$) of all fittings indicates that the model curves were reliable. The hydrolysis constant (k) can reflect the hydrolysis rate of the digested substrate, and the higher the p value, the better the hydrolysis property (Chen, Zhu, and Jiang et al. 2019b). It is evident that the hydrolysis rate of the substrate was increased by MAFC pretreatment (Table 5). In addition, the lag-phase time (λ) of MAFC increased by 34.29% compared with CK (1.40). Longer λ indicates slower adaptation to the anaerobic digestion as well as lower CMP

Table 5. First-order kinetic model and modified Gompertz model fitting results.

Groups	First-order kinetic model			Modified Gompertz model			
	B_m	k	R^2	G_m	r_m	λ	R^2
	mL /g VS	day^{-1}	–	mL/g VS	mL/g VS/day	day	–
CK	116.51 ± 5.67	0.09 ± 0.01	0.970	97.00 ± 0.65	9.13 ± 0.23	1.40 ± 0.13	0.997
MAFC	130.19 ± 6.42	0.11 ± 0.01	0.945	113.78 ± 0.28	14.11 ± 0.17	1.88 ± 0.05	0.999

(Wang, Liu, and Noor et al. 2019b). This may be because the methanogens needed more lag phase time to transition to the new environment changed by MAFC, which was consistent with the trend of daily methane production at the beginning of anaerobic digestion. For the simulated CMP, the predicted results of the modified Gompertz model were more similar to the actual CMP. However, although the CMP predicted by the two models was different, both showed higher methane production in the MAFC pretreatment group.

Analysis of microbial community composition and diversity

Anaerobic digestion is a multistage biochemical process in which many kinds of microorganisms act together. The influence of MAFC pretreatment on anaerobic digestion performance can be analyzed by the changes of microbial diversity and community at different times (Sepehri and Sarrafzadeh 2018). Therefore, the diversity and structure were evaluated in the 5th, 15th, and 25th days of anaerobic digestion.

Effects of MAFC pretreatment on microbial diversity

Alpha diversity reflects the species diversity of a single sample. The ACE and Chao1 index are used to estimate the total number of species in the sample, and the higher the value, the more species. Shannon and Simpson indexes are commonly used to reflect the diversity of alpha. The higher the Shannon value is, the higher the community diversity is, while the Simpson value is the opposite. From Table 6, the sequence numbers of CK were 45221–72453 for bacteria and 48967–55216 for archaea, with a total of 1143–1782 and 532–631 OTUs detected for bacteria and archaea at day 5, 15, and 25, respectively. Whereas the sequence numbers of MAFC were 48346–72950 for bacteria and 56085–58822 for archaea, with a total of 1031–2113 and 597–683 OTUs detected for bacteria and archaea, respectively. Higher Alpha diversity reflects better ecological stability and higher toxicity resistance during anaerobic digestion (Lu, Xing, and Ren 2012). Therefore, the application of MAFC pretreatment improved the stability of the anaerobic digestion system. In addition, the changes in ACE, Chao1, Shannon, and Simpson indexes also confirmed that MAFC was beneficial to the stability of anaerobic digestion.

Table 6. Alpha diversity analysis for bacteria and archaea.

Time	Group	Sequence Number	OUT number	Shannon index	Simpson index	ACE index	Chao1 index
5	CK_b	45221	1143	3.52	0.11	2311.43	1873.65
	$MAFC_b$	48346	1440	4.47	0.03	3458.57	2511.13
	CK_a	48967	532	2.10	0.23	2655.88	1434.81
	$MAFC_a$	56085	597	2.22	0.20	3312.31	2097.06
15	CK_b	68972	1765	3.94	0.18	3804.10	2624.34
	$MAFC_b$	58666	1031	3.83	0.28	2381.37	1812.51
	CK_a	55216	631	2.28	0.15	3490.97	2713.11
	$MAFC_a$	58822	683	2.20	0.20	3796.78	2716.44
25	CK_b	72453	1782	2.69	0.26	3906.18	2874.52
	$MAFC_b$	72950	2113	2.35	0.32	2899.34	2138.64
	CK_a	54372	614	3.07	0.13	3695.25	2834.16
	$MAFC_a$	57782	673	2.43	0.16	3803.51	3186.55

Note: $_B$ is for bacteria; $_a$ is for Archaea.

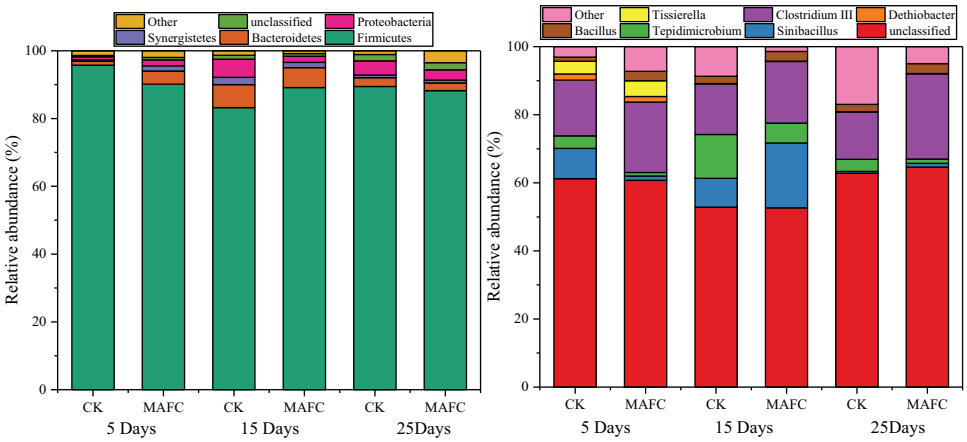


Figure 4. Relative abundance of bacteria at phylum (a) and genus level (b).

Effects of MAFC pretreatment on microbial community

As shown in Figure 4(a), four dominant phyla, namely *Firmicutes, Bacteroidetes, Synergistetes*, and *Proteobacteria*, were identified in anaerobic digestion. *Firmicutes* accounted for the largest proportion in the whole process of anaerobic digestion that mainly released lipases, proteases, and cellulases that hydrolyze organic matter (Cai, Hua, and Gao et al. 2017). *Bacteroidetes* dominates the conversion of carbohydrates, fats, and proteins to VFA (Wang, Zhang, and Zhou et al. 2017). *Synergistetes* and *Proteobacteria* play a leading role in the conversion of VFA to acetic acid/hydrogen, which can be used to feed the aceticlastic and hydrogenotrophic methanogens (Jiang, Li, and Li et al. 2020).

On the 5th day of anaerobic digestion, the relative abundances of *Firmicutes, Bacteroidetes, Synergistetes*, and *Proteobacteria* in CK were 95.73%, 1.15%, 1.71%, and 0.89%, respectively. In contrast, only the relative abundance of *Firmicutes* decreased in MAFC (90.17%), while the relative abundance of *Bacteroidetes, Synergistetes*, and *Proteobacteria* increased by 234.78%, 248.89%, and 92.13%, respectively. Previous reports have shown that MA provided a suitable oxygen environment to accelerate the growth of facultative bacteria, which was conducive to their absorption of small molecule compounds to release VFA (Charles, Walker, and Cord-Ruwisch 2009; Sun, Zhang, and Sun et al. 2020b). While, Yu, Shan, and Zhang et al. (2015b) also found that the addition of FC could inhibit excessive VFA synthesis and maintain system stability. Thus, MAFC pretreatment could balance the stability of the system while accelerating the acidogenic phase, that is, the conversion of small molecule compounds to VFA and further acetic acid/hydrogen. On the 15th day, the relative abundance of *Firmicutes* pretreated by MAFC was 7.10% higher than that of CK, while the other three phyla were lower than that of CK, indicating that the acidogenic phase was weakened for the MAFC group compared with CK. This result is supported by the above trends in daily methane production. On the 25th day, there was little difference in relative abundance of the digestive system before and after MAFC pretreatment on the bacterial phylum level.

Figure 4(b) depicts the relative abundance of bacterial communities at genus level on the 5th day. Six genera with high relative abundances were individually figured, and they were *Sinibacillus, Tepidimicrobium, Clostridium* III, *Dethiobacter*, *Tisierella*, and *Bacillus. Clostridium* was the most abundant genus of bacteria for CK and MAFC with relative abundances of 16.36% and 20.66%, respectively. It is a typical hydrolytic and acidogenic genus of bacteria in thermophilic anaerobic digestion. Also, a higher relative abundance of *Clostridium* for MAFC meant a better rate of hydrolysis during the prophase of digestion (Ziganshin, Liebetrau, and Pr?Ter 2013). The relative abundance of

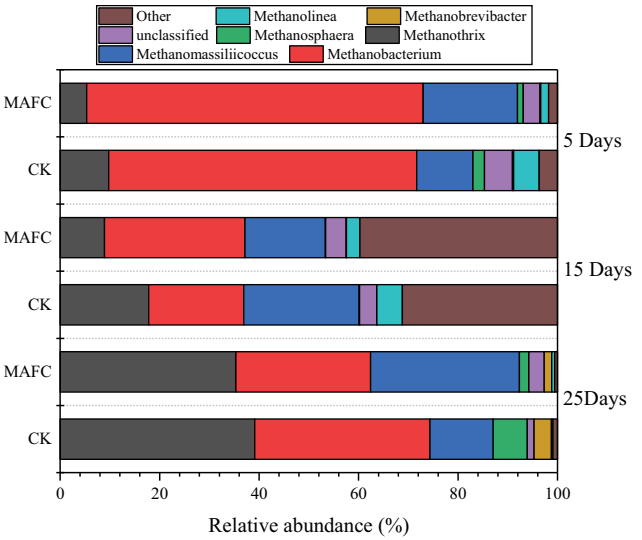


Figure 5. Relative abundance of archaea at the genus level.

Sinibacillus significantly increased after MAFC pretreatment on day 15, while that of CK changed little. The increase of the relative abundance of *Sinibacillus*, a facultative thermophilic bacterium with the function of degrading organic matter, might be caused by early MA. In addition, the proportion of *Tepidimicrobium* is seen as an indicator of the acidification process, its relative abundance in MAFC was lower than that in CK throughout anaerobic digestion, suggesting that the former has a more stable digestive system (Regueiro, Lema, and Carballa 2015).

Several major archaea were identified and listed in Figure 5. There were five hydrogenotrophic methanogens, namely, *Methanobacterium, Methanomassiliicoccus, Methanosphaera, Methanobrevibacter*, and *Methanolinea*, which metabolize hydrogen and carbon dioxide to produce methane (Capson-Tojo, Moscoviz, and Astals et al. 2020). As the only acetoclastic archaea in this study, *Methanothrix* played a dominant role, and its relative abundance increased with time. Dissimilarly, the relative abundances of *Methanothrix* in MAFC were 9.66%, 49.94%, and 45.12% lower than that of CK on day 5, 15, and 25, respectively. The results showed that MAFC pretreatment was not suitable for the growth and reproduction of *Methanothrix*. It has been reported that acetoclastic methanogens were more sensitive to environmental changes than hydrogenotrophic methanogens (Lü, Hao, and Guan et al. 2013). Thus, MAFC pretreatment enhanced methanogenesis by enhancing the hydrogenotrophic pathway.

Conclusions

This study demonstrated that MAFC pretreatment promoted the methanogenic properties of thermophilic dry anaerobic digestion. Both MA and FC had significant effects on CMP, and the optimal pretreatment parameters were: oxygen loading rate of 0.76 mL/g VS and FC addition rate of 4.04 mg/g. The CMP of MAFC pretreatment was 114.05 mL/g VS with the optimal conditions, which was 17.78% higher than that of CK. In addition, microbial community analysis showed that MAFC pretreatment enhanced methanogenesis by enhancing the hydrogenotrophic pathway. The success of this pretreatment provides a potential way to improve the performance of dry anaerobic digestion.

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Disclosure statement

No potential conflict of interest was reported by the author(s).

References

Ahmadi, M., F. Vahabzadeh, B. Bonakdarpour, Mofarrah, E., and Mehranian, M. . 2005. Application of the central composite design and response surface methodology to the advanced treatment of olive oil processing wastewater using Fenton's peroxidation. *Journal of Hazardous Materials* 123 (1):187–95.

Amin, F. R., H. Khalid, W. Li, Chen, C., and Liu, G. . 2021. Enhanced methane production and energy potential from rice straw by employing microaerobic pretreatment via anaerobic digestion. *Journal of Cleaner Production* 296:126434.

Baek, G., J. Kim, and C. Lee. 2014. Influence of ferric oxyhydroxide addition on biomethanation of waste activated sludge in a continuous reactor. *Bioresource Technology* 166:596–601.

Brémond, U., R. de Buyer, J.-P. Steyer, Bernet, N., and Carrere, H. . 2018. Biological pretreatments of biomass for improving biogas production: An overview from lab scale to full-scale. *Renewable and Sustainable Energy Reviews* 90:583–604.

Cai, Y., B. Hua, L. Gao, Hu, Y., Yuan, X., Cui, Z., Zhu, W., and Wang, X. . 2017. Effects of adding trace elements on rice straw anaerobic mono-digestion: Focus on changes in microbial communities using high-throughput sequencing. *Bioresource Technology* 239:454–63.

Capson-Tojo, G., R. Moscoviz, S. Astals, Robles, Á., and Steyer, J. P. . 2020. Unraveling the literature chaos around free ammonia inhibition in anaerobic digestion. *Renewable and Sustainable Energy Reviews* 117:109487.

Charles, W., L. Walker, and R. Cord-Ruwisch. 2009. Effect of pre-aeration and inoculum on the start-up of batch thermophilic anaerobic digestion of municipal solid waste. *Bioresource Technology* 100 (8):2329–35.

Chen, Y., H. Yang, Z. Zhao, Zou, H., Zhu, R., Jiang, Q., Sun, T., Li, M., Li, L., Shi, D., Ai, H., He, Q., and Gu, L. . 2019a. Comprehensively evaluating the digestive performance of sludge with different lignocellulosic components in mesophilic anaerobic digester. *Bioresource Technology* 293:122042.

Chen, Y., R. Zhu, Q. Jiang, Sun, T., Li, M., Shi, J., Chai, H., Gu, L., Ai, H., and He, Q. . 2019b. Effects of green waste participation on the co-digestion of residual sludge and kitchen waste: A preliminary study. *Science of the Total Environment* 671:838–49.

Deepanraj, B., N. Senthilkumar, and J. Ranjitha. 2021. Effect of solid concentration on biogas production through anaerobic digestion of rapeseed oil cake. *Energy Sources, Part A: Recovery, Utilization, and Environmental Effects* 43 (11):1329–36.

Esen, M., and T. Yuksel. 2013. Experimental evaluation of using various renewable energy sources for heating a greenhouse. *Energy and Buildings* 65:340–51.

Fu, S.-F., F. Wang, X.-Z. Yuan, Yang, Z.-M., Luo, S.-J., Wang, C.-S., and Guo, R.-B. . 2015. The thermophilic (55°C) microaerobic pretreatment of corn straw for anaerobic digestion. *Bioresource Technology* 175:203–08.

Gunes, B., J. Stokes, P. Davis, Connolly, C., and Lawler, J. . 2021. Optimisation of anaerobic digestion of pot ale after thermochemical pre-treatment through Response Surface Methodology. *Biomass and Bioenergy* 144:105902.

Jiang, J., L. Li, Y. Li, He, Y., Wang, C., and Sun, Y. . 2020. Bioaugmentation to enhance anaerobic digestion of food waste: Dosage, frequency and economic analysis. *Bioresource Technology* 307:123256.

Kainthola, J., A. S. Kalamdhad, and V. V. Goud. 2019a. Optimization of methane production during anaerobic co-digestion of rice straw and hydrilla verticillata using response surface methodology. *Fuel* 235:92–99.

Kainthola, J., A. S. Kalamdhad, and V. V. Goud. 2019b. A review on enhanced biogas production from anaerobic digestion of lignocellulosic biomass by different enhancement techniques. *Process Biochemistry* 84:81–90.

Lim, J. W., and J.-Y. Wang. 2013. Enhanced hydrolysis and methane yield by applying microaeration pretreatment to the anaerobic co-digestion of brown water and food waste. *Waste Management* 33 (4):813–19.

Lü, F., L. Hao, D. Guan, Qi, Y., Shao, L., and He, P. . 2013. Synergetic stress of acids and ammonium on the shift in the methanogenic pathways during thermophilic anaerobic digestion of organics. *Water Research* 47 (7):2297–306.

Lu, L., D. Xing, and N. Ren. 2012. Pyrosequencing reveals highly diverse microbial communities in microbial electrolysis cells involved in enhanced H2 production from waste activated sludge. *Water Research* 46 (7):2425–34.

Nie, E., P. He, H. Zhang, Hao, L., Shao, L., and Lü, F. . 2021. How does temperature regulate anaerobic digestion? *Renewable and Sustainable Energy Reviews* 150:111453.

Qin, Y., L. Chen, T. Wang, Ren, J., Cao, Y., and Zhou, S. . 2019. Impacts of ferric chloride, ferrous chloride and solid retention time on the methane-producing and physicochemical characterization in high-solids sludge anaerobic digestion. *Renewable Energy* 139:1290–98.

Regueiro, L., J. M. Lema, and M. Carballa. 2015. Key microbial communities steering the functioning of anaerobic digesters during hydraulic and organic overloading shocks. *Bioresource Technology* 197:208–16.

Rocamora, I., S. T. Wagland, R. Villa, Simpson, E. W., Fernández, O., and Bajón-Fernández, Y., et al. 2020. Dry anaerobic digestion of organic waste: A review of operational parameters and their impact on process performance. *Bioresource Technology* 299:122681.

Sepehri, A., and M.-H. Sarrafzadeh. 2018. Effect of nitrifiers community on fouling mitigation and nitrification efficiency in a membrane bioreactor. *Chemical Engineering and Processing - Process Intensification* 128:10–18.

Sheets, J. P., X. Ge, and Y. Li. 2015. Effect of limited air exposure and comparative performance between thermophilic and mesophilic solid-state anaerobic digestion of switchgrass. *Bioresource Technology* 180:296–303.

Sun, Y., Z. Wang, Y. Liu, Meng, X., Qu, J., Liu, C., and Qu, B. . 2020a. A Review on the Transformation of Furfural Residue for Value-Added Products. *Energies* 13(1):21.

Sun, Y., Z. Zhang, Y. Sun, and Yang, G. . 2020b. One-pot pyrolysis route to FeN-Doped carbon nanosheets with outstanding electrochemical performance as cathode materials for microbial fuel cell. *International Journal of Agricultural and Biological Engineering* 13 (6):8.

Ugwu, S. N., and C. C. Enweremadu. 2020. Enhancement of biogas production process from biomass wastes using iron-based additives: Types, impacts, and implications. *Energy Sources Part A: Recovery, Utilization, and Environmental Effects*, 1-23.

Wang, Z., Q. Cheng, Z. Liu, Qu, J., Chu, X., Li, N., Noor, R. S., Liu, C., Qu, B., and Sun, Y. . 2019a. Evaluation of methane production and energy conversion from corn stalk using furfural wastewater pretreatment for whole slurry anaerobic co-digestion. *Bioresource Technology* 293:121962.

Wang, Z., J. Li, F. Yu, Yan, B., and Chen, G., . 2022. Comprehensive evaluation of gradient controlled anaerobic digestion and pyrolysis integration processes: A case study of Sargassum treatment. *Bioresource Technology* 345:126496.

Wang, Z., Z. Liu, R. S. Noor, Cheng, Q., Chu, X., Qu, B., Zhen, F., and Sun, Y. . 2019b. Furfural wastewater pretreatment of corn stalk for whole slurry anaerobic co-digestion to improve methane production. *Science of the Total Environment* 674:49–57.

Wang, M., X. Zhang, J. Zhou, Yuan, Y., Li, D., Li, Z., Liu, X., and Yan, Z. . 2017. The dynamic changes and interactional networks of prokaryotic community between co-digestion and mono-digestions of corn stalk and pig manure. *Bioresource Technology* 225:23–33.

Xu, Y., M. K. Awasthi, P. Li, Meng, X., and Wang, Z. . 2020. Comparative analysis of prediction models for methane potential based on spent edible fungus substrate. *Bioresource Technology* 317:124052.

Xu, S., A. Selvam, and J. W. C. Wong. 2014. Optimization of micro-aeration intensity in acidogenic reactor of a two-phase anaerobic digester treating food waste. *Waste Management* 34 (2):363–69.

Yu, B., Z. Lou, D. Zhang, Shan, A., Yuan, H., Zhu, N., and Zhang, K. . 2015a. Variations of organic matters and microbial community in thermophilic anaerobic digestion of waste activated sludge with the addition of ferric salts. *Bioresource Technology* 179:291–98.

Yu, B., A. Shan, D. Zhang, Lou, Z., Yuan, H., Huang, X., Zhu, N., and Hu, X. . 2015b. Dosing time of ferric chloride to disinhibit the excessive volatile fatty acids in sludge thermophilic anaerobic digestion system. *Bioresource Technology* 189:154–61.

Yu, B., D. Zhang, X. Dai, Lou, Z., Yuan, H., and Zhu, N. . 2016. The synthetic effect on volatile fatty acid disinhibition and methane production enhancement by dosing FeCl3 in a sludge thermophilic anaerobic digestion system. *RSC Advances* 6 (25):21090–98.

Zhan, W., Y. Tian, J. Zhang, Zuo, W., Li, L., Jin, Y., Lei, Y., Xie, A., and Zhang, X. . 2021. Mechanistic insights into the roles of ferric chloride on methane production in anaerobic digestion of waste activated sludge. *Journal of Cleaner Production* 296:126527.

Zhang, W., S. Wu, J. Guo, Zhou, J., and Dong, R. . 2015. Performance and kinetic evaluation of semi-continuously fed anaerobic digesters treating food waste: Role of trace elements. *Bioresource Technology* 178:297–305.

Zhao, Z., Y. Li, Z. Zhao, Yu, Q., and Zhang, Y. . 2020. Effects of dissimilatory iron reduction on acetate production from the anaerobic fermentation of waste activated sludge under alkaline conditions. *Environmental Research* 182:109045.

Zhen, F., X. Luo, T. Xing, Sun, Y., Kong, X., and Li., W. . 2021. Performance evaluation and microbial community analysis of microaerobic pretreatment on thermophilic dry anaerobic digestion. *Biochemical Engineering Journal* 167:107873.

Ziganshin, A. M., J. Liebetrau, and J. Pr?Ter. 2013. Microbial community structure and dynamics during anaerobic digestion of various agricultural waste materials. *Applied Microbiology & Biotechnology* 97 (11):5161–74.